

ME5627 Measurements in Thermofluids

Laboratory Record

Submitted By

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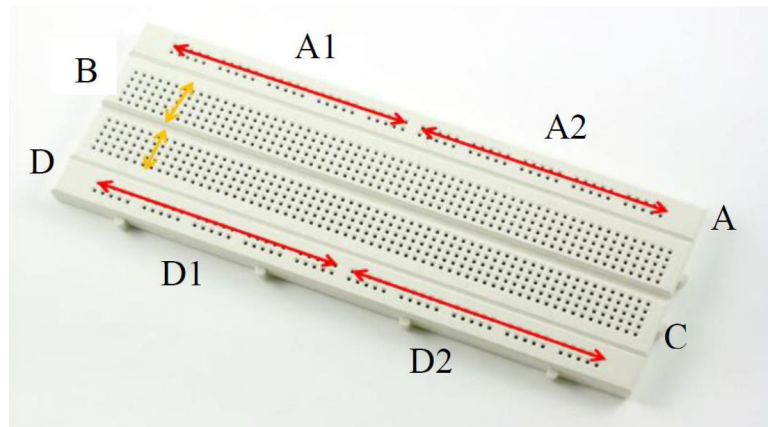
EXPERIMENT 1

PART A: FAMILIARISATION WITH PC SCOPE AND BREAD BOARD THROUGH RC CIRCUIT

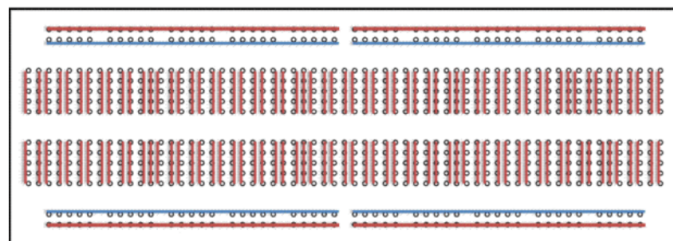
Aim – To gain familiarity with Bread Board, oscilloscope and oscilloscope data logger through RC circuit

Apparatus – Bread board, wires, Oscilloscope, Resistor, capacitor

Details of Bread Board:



(a) Schematic of the Bread Board



(b) Equipotential lines

Figure 1.1: Breadboard familiarisation

- Hold the breadboard such that the horizontal side is longer than the vertical side.
- There are four independent portions (A, B, C and D) created by three notches as in Fig. 1.1a. Here, independence means portions A, B, C and D are not connected with each other

Portion A and D

- Each portion of A and D is further subdivided into two regions as A1 and A2 shown in Fig. 1.1.
- Each horizontal line (shown through arrows in the Fig. 1.1a) of A1, A2, D1 and D2 represent a single potential. Usually, as a general practice, these points are considered as ground for a given built circuit.

Portion B and C

- In portion B and C, each vertical line (shown through arrows in Fig. 1.1a) of B and C represents a single potential.
- The above explanation can be better understood through Fig. 1.1b showing same potential lines.

Details of oscilloscope probe

Fig. 1.2 shows a typical oscilloscope probe.

- The Oscilloscope probe is a passive connector which is used to connect oscilloscope to the electrical circuit. It consists of three parts Retractable hook tip (or crocodile clip), a Crocodile clip and BNC connector.
- BNC connector: It is used to connect oscilloscope to the probe.
- Retractable Hook tip: It is used to connect the node of the electrical circuit to the oscilloscope where there is a need to measure the signal.
- Crocodile clip: It is connected to the ground of the electrical circuit in the bread board.
- Make sure the RED slider on the probe should be on the X1 position only. X1 means amplification is unity.



Figure 1.2: PC scope probe

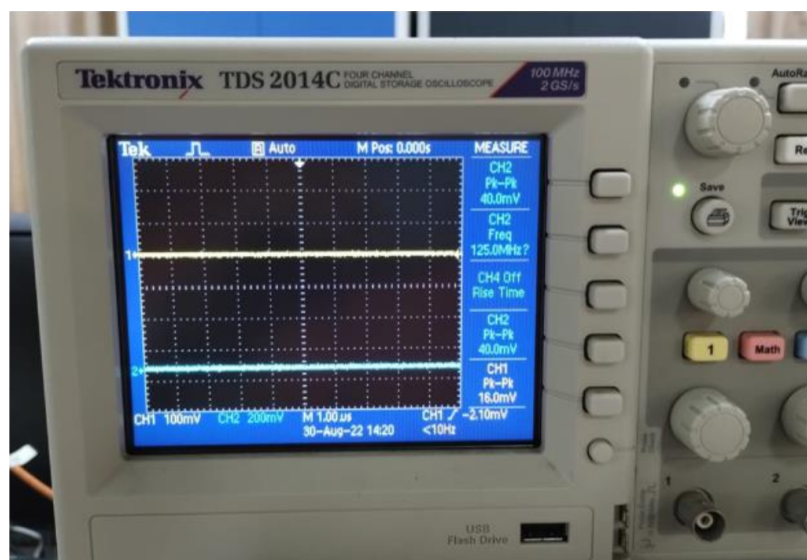


Figure 1.3: Four channel Oscilloscope with digital storage

Fig. 1.3 shows a four channel Oscilloscope with digital data storage system. There are four channels which displays the input/output signal. Input and output probes may be connected to any of these four channels to visualize and analyze the waveforms of the signal. There are scaling, vertical and horizontal controls to adjust the waveform

in order to have better visualization. Peak to peak voltage and time scale can be conveniently read from the monitor using the adjustment knob after pressing the cursor button.



Figure 1.4: Function Generator

In order to obtain a desired input signal (sinusoidal, square, ramp, pulse or any arbitrary waveform) a function generator can be used (Fig. 1.4). The desired amplitude and frequency of the waveform can be obtained using the side controls near the monitor. The output probes may be connected to any of the two channels. One of the crocodile clip of the output probe needs to be grounded and the other end can be given as input signal to generate the signal in the circuit. Also, in order to check whether the function generator is working, the probe may be connected to the Oscilloscope.

Calculation of Resistance:

- Turn the resistor so that the gold or silver stripe is at the right end of the resistor.
- Look at the color of the first two stripes on the left end. These correspond to the first two digits of the resistor value. Use the table given below to determine the first two digits.
- Look at the third stripe from the left. This corresponds to a multiplication value. Find the value using the table below.
- Multiply the two digit number from step two by the number obtained from step three. This is the value of the resistor in ohms. The fourth stripe indicates the accuracy of the resistor.

Sample:

You are given a resistor whose stripes are colored from left to right as brown, black, orange, gold. Find the resistance value.

- The gold stripe is on the right so go to Step Two.
- The first stripe is brown which has a value of 1. The second stripe is black which has a value of 0. Therefore the first two digits of the resistance value are 10.
- The third stripe is orange which means $\times 1,000$.
- The value of the resistance is found as $10 \times 1000 = 10,000$ ohms i.e., $10 \text{ k}\Omega$
- The gold stripe means the actual value of the resistor may vary by 5% meaning the actual value will be somewhere between 9,500 ohms and 10,500 ohms. (Since $5\% \text{ of } 10,000 = 0.05 \times 10,000 = 500$)

Table 1.1: Resistor Color Code Table

Color	Numeric Value of 1st Band	Numeric Value of 2nd Band	3rd Band as Multiplier	4th Band as Tolerance
Black	0	0	$\times 10^0$	-
Brown	1	1	$\times 10^1$	-
Red	2	2	$\times 10^2$	-
Orange	3	3	$\times 10^3$	-
Yellow	4	4	$\times 10^4$	-
Green	5	5	$\times 10^5$	-
Blue	6	6	$\times 10^6$	-
Violet	7	7	$\times 10^7$	-
Gray	8	8	$\times 10^8$	-
White	9	9	$\times 10^9$	-
Gold	-	-	$\times 10^{-1}$	$\pm 5\%$
Silver	-	-	$\times 10^{-2}$	$\pm 10\%$
No Color	-	-	-	$\pm 20\%$

Calculation of Capacitance:

Sample Calculation:

- Step 1: In the Fig. 1.5, the first two digits from the left indicates the first two digits of the capacitor value. Here, in this case the first two digits are 1 and 0. Therefore, the first two digits of the capacitor value is 10.
- Step 2: The third digit is 4 which means that four zeroes would be followed by 1 (10,000 pF).
- The value of the capacitor is found out to be the product of the number obtained in step 1 and step 2. That is, $10 \times 10000 = 10,0000 \text{ pF}$ i.e $0.1 \mu\text{F}$.

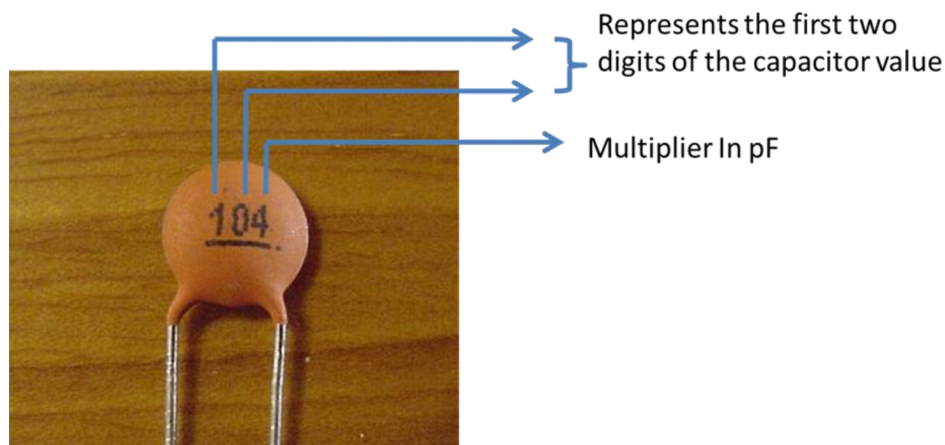


Figure 1.5: Ceramic capacitor

Circuit Diagram:

Figures. 1.6 and 1.7 show the RC circuit and PC scope

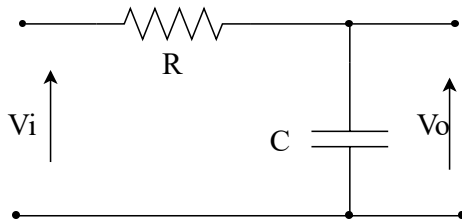


Figure 1.6: Typical RC circuit.



Figure 1.7: PC Scope.

Experimental Procedure:

- Build an RC circuit as shown in the Fig. 1.6 on the bread board.
- **Circuit input to the oscilloscope**
First PC scope probe is taken. BNC connector of this oscilloscope is connected to the input of the oscilloscope shown in Fig. 1.7. Connect the retractable hook tip of the oscilloscope probe (input probe) to the open end of the resistor. The crocodile clip of this input PC scope probe is connected to the ground of the bread board.
- **Circuit Output to the oscilloscope**
Second oscilloscope probe is taken. BNC connector of this oscilloscope is connected to the output of the scope shown in Fig. 1.7. Connect the retractable hook tip of the scope probe (output probe) to common junction of resistor and capacitor. The crocodile clip of this input oscilloscope probe is connected to the ground of the bread board.
- **Desired Signal Input to the oscilloscope**
Third oscilloscope probe is taken. BNC connector of this oscilloscope is connected to the signal generator of the oscilloscope shown in Fig. 1.7. Connect the retractable hook tip of the oscilloscope probe (input probe) to the open end of the resistor. The crocodile clip of this input oscilloscope probe is connected to the ground of the bread board.

Following flow chart shows the flow of input and output signals.

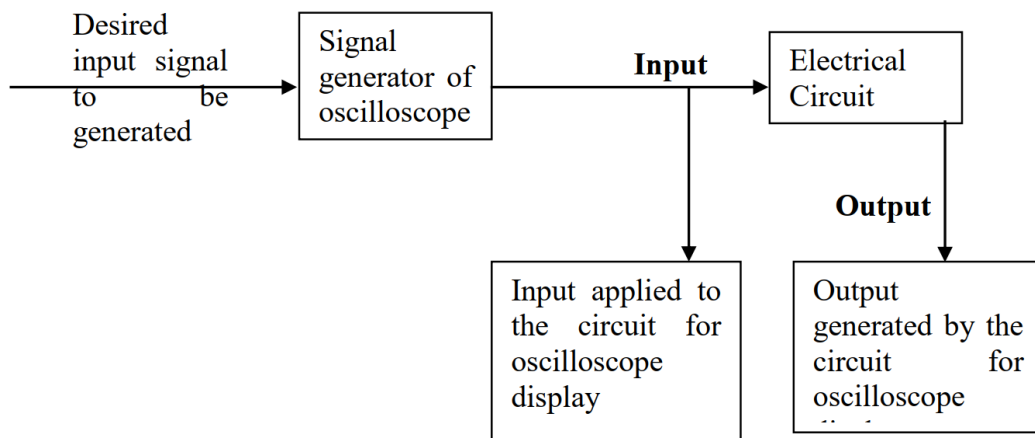


Figure 1.8: Flowchart indicating the flow of input and output signals

- There is switch on the probe. Make sure that switch is towards 'X1' side so that the values are not amplified by the probe.
- Give sinusoidal signal of very low amplitude (2 V) as input and observe the output.
- Increase the amplitude of the input signal and observe the output, repeat this step three times.
- Measure the peak voltage of both input and output signal using meter A and B.
- Try giving some other type of signal at some other frequency and observe/measure the output.

Conclusions/Discussions on the results:

Table 1.2: Resistor value determination.

Sr.No	Band	Color Code	Numerical Value
1	1 st	Brown	1
2	2 nd	Green	5
3	3 rd	Brown	10 ¹
4	4 th	Gold	5%

From Table. 1.2, the resistance value can be determined. The resistance value is $150\Omega \pm 5\%$

- Understanding the resistance color coding and determine the stated value of a resistor by interpreting the color code indicated on the resistor and verify the value of resistance using millimeter
- Gained an understanding of resistance color coding and learned how to get the nominal value of a resistor by interpreting the color bands marked on it.
- Verified the accuracy of the resistor value by measuring it with a multi meter, compared the measured resistance with the value predicted from the color code
- Gained familiarity with the bread board and various connection pattern used in circuit assembly. This involved the understanding the layout and how different components are interconnected on the bread board.
- Employed four channel oscilloscope and function generator for the experimentation .Detailed observation were made regarding the operation and behavior of these equipment, noting how they interact with the circuit under test
- Conducted an in-depth study of RC circuit to understand its various response characteristics
- When the output is taken from the capacitor side the RC circuit, the circuit behaves as low pass filter. This means it allows low frequency signals to pass through while attenuating higher frequency signals
- In contrast the output is taken from the resistance side, the circuit function as high pass filter, which allows high frequency signals to pass while blocking lower frequency signal.

EXPERIMENT 1

PART B: BASIC WORKING OF INSTRUMENTATION OP AMP (INA 129)

Aim – To study the basic working of Instrumentation op amp (INA129)

Apparatus – INA 129, Bread board, wires, dc power supply, Oscilloscope

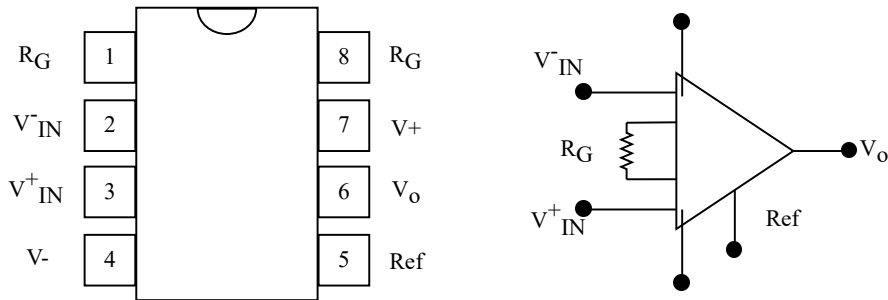


Figure 1.9: Schematic of the instrumentation operational amplifier with pins (INA 129)

Experimental Procedure

- Make the circuit as shown in the Fig. 1.8 on the bread board.
- Connect the Retractable hook tip of the input probe and signal generator probe of the oscilloscope at pin 3 and the crocodile clip of both the probe (input and signal generator) at pin 2 of INA 129.
- Connect the Retractable hook tip of the output probe at pin 6 and the crocodile clip of the output probe at pin 5 of INA 129. Make a virtual ground in the breadboard and connect pin 5 to this virtual ground.
- Select the value of R_G based on desired gain. The relation between R_G and gain/Amplification (G) is given in Table. 1.3
- Connect the resistance R_G across pin 1 and 8 of INA 129.
- D.C power supply is set to 12V whose positive end is connected to pin 7 while its negative end is connected to pin 4.
- Give sinusoidal signal of 200mV amplitude and the frequency of 300 Hz in the signal generator and observe the output at the interface of oscilloscope in the monitor. Ensure that pin 5 is grounded.
- Change the value of R_G , repeat the above procedure and observe the output.
- Note down the output in the observation Table.

Table 1.3: Relation between R_G and Desired gain

Sr.No	R_G ($k\Omega$)	Selected Input (mV)	Measured Output (mV)	Measured Gain (G)	Calculated Gain
1	49.4	200	408	2.04	2
2	12.35	200	1060	5.3	5
3	1.008	200	10200	51	50

Sample calculation for set 2:

- The relation between desired gain and R_G is given by

$$G = 1 + \frac{49.4 \text{ k}\Omega}{R_G} \quad (1.1)$$

The R_G is the resistor in an operational amplifier that sets the amplifier's overall gain. In this, our desired gain is 5. So from Eq. 1.1, we can calculate $R_G = 12.35 \text{ k}\Omega$.

- Measured gain (G) is calculated by

$$G = \frac{\text{Measured output}}{\text{Selected input}} = \frac{1060}{200} = 5.3 \quad (1.2)$$

Table 1.4: Desired Gain and Corresponding R_G Values

Desired Gain (V/V)	R_G (Ω)	Nearest 1% R_G (Ω)
1	NC	NC
2	49.4K	49.9K
5	12.35K	12.4K
10	5489	5.49K
20	2600	2.61K
50	1008	1K
100	499	499
200	248	249
500	99	100
1000	49.5	49.9
2000	24.7	24.9
5000	9.88	9.76
10000	4.94	4.87

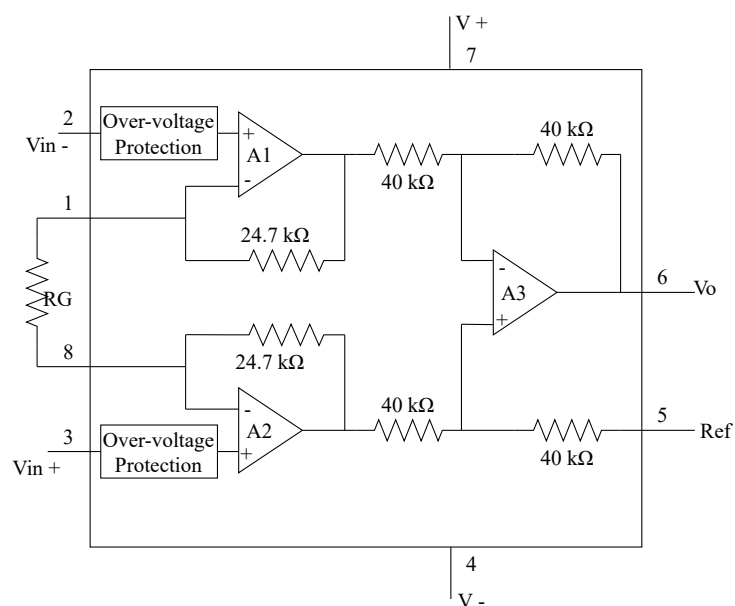


Figure 1.10: Internal circuit of Op-Amp

Conclusions/Discussion on the Results:

- The operation of the INA 129 instrumentation op-amp was thoroughly observed and analyzed, demonstrating its effectiveness in amplifying small input signals with high precision.
- The gain was measured across various resistor values, showing that the experimental results closely matched the theoretical values.
- The INA 129 is designed for high-precision applications, offering excellent common-mode rejection and high input impedance, making it ideal for amplifying low-level signals in noisy environments.
- The input DC voltage significantly impacts the output signal, as it can shift the entire waveform vertically and potentially drive the output into saturation if the DC offset is too high, highlighting the importance of managing DC offsets in practical applications.
- The gain, determined by the resistor R_G connected between pins 1 and 8, varied with different resistor values, illustrating the versatility of the INA 129. A measured output of 1.06 mV with R_G at 12.35 k Ω resulted in a gain of approximately 5.3, closely aligning with the theoretical gain of 5. Similarly, adjusting R_G to 48.4 k Ω resulted in a lower gain, consistent with expected amplifier behavior. Overall, the measured gains consistently matched the calculated values, confirming the reliability of the INA 129.
- This amplifier is particularly suitable for applications requiring high accuracy and low noise, such as medical instrumentation and sensor data acquisition, where preserving signal integrity is critical.
- The experiment provided valuable insights into the behavior of instrumentation amplifiers and their role in enhancing signal fidelity in complex electronic systems, suggesting that future investigations could further explore the INA 129's performance under various input types and conditions to assess its robustness and versatility.

EXPERIMENT 2

DYNAMIC RESPONSE OF A FIRST ORDER SYSTEM USING R-C CIRCUIT

Aim

- To study and understand the behavior of a first order system simulated by an RC circuit using an oscilloscope-cum-signal generator with sinusoidal and ramp inputs. The output amplitude and the phase difference are studied with various frequencies of the input signal.
- Find the response characteristic of RC circuit for a generated square wave using Fourier series as input compare with the theoretical response

Apparatus – Resistor, capacitor, breadboard, oscilloscope

Procedure:

- Construct the circuit as shown in Fig. 2.1 with given electronic components.

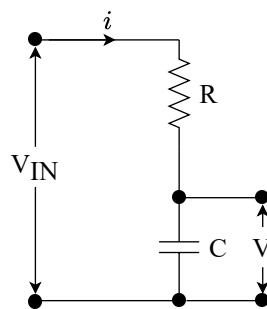


Figure 2.1: R-C Circuit

- Chose appropriate resistance and capacitance ($R = 1k\Omega$ and $C = 0.1\mu F$). Calculate the cut off frequency $f = \frac{1}{2\pi RC}$
- Connect the oscilloscope-cum-signal generator output probe to supply the input as shown. Connect “Channel A” probes of the oscilloscope-cum-signal generator to the input probe and “Channel B” across the capacitor to extract the output.
- The input to the circuit will be sinusoidal and square signals of constant amplitude (5 volts peak to peak) and variable frequency in order to obtain the frequency response characteristics.
- Apply a sinusoidal wave as the input to the RC circuit. Study the output across the capacitor on the screen and make quantitative observations (see Table. 2.1) regarding the amplitude and phase of input and output waveform.
- Using the oscilloscope function in the PC software, Export the input and output signals to a “.csv” file for further calculation. Measure the amplitude and phase difference between input and output in MS Excel.
- Take three different readings above, three readings below and one at the cut off frequency.
- Calculate the static gain in each case, plot the frequency response of the given RC circuit for the static gain, and phase difference on the graph sheets.

- Repeat the experiment for square wave as the input to the RC circuit. However, select input frequencies such that they are lower than the cut off frequency.
- Plot decibel value variation with frequency (Bode Plot) and measure the slope

$$\text{Gain in dB} = 20 \log \left| \frac{V_o}{V_i} \right| \quad (2.3)$$

- Take the input signal and output signal (generated from the circuit) from the generated .csv file. Calculate the output signal using the frequency response characteristics of first order system.

Observations:

Resistance $R = 1 \text{ k}\Omega$

Capacitance $C = 0.1 \text{ }\mu\text{F}$

Table 2.1: Variation of output voltage with frequency for sinusoidal input.

Sr. No.	Frequency (Hz)	Input peak to peak V_i (Volts)	Output peak to peak V_o (Volts)	Measured static gain $\left(\frac{V_o}{V_i}\right)$	Calculated static gain	Measured Phase difference ϕ (rad)	Calculated phase difference (rad)
1	200	1.15	1.150	1.000	0.992	-0.125	-0.125
2	400	1.15	1.110	0.965	0.970	-0.279	-0.246
3	600	1.15	1.050	0.913	0.936	-0.419	-0.361
4	800	1.15	0.980	0.852	0.893	-0.559	-0.466
5	1000	1.15	0.920	0.800	0.847	-0.628	-0.561
6	1200	1.15	0.880	0.765	0.798	-0.698	-0.646
7	1400	1.15	0.820	0.713	0.751	-0.768	-0.721
8	1500	1.15	0.800	0.696	0.728	-0.785	-0.756
9	1600	1.15	0.750	0.652	0.705	-0.873	-0.788
10	1800	1.15	0.719	0.625	0.662	-0.925	-0.847
11	2000	1.15	0.700	0.609	0.623	-0.960	-0.899
12	2200	1.15	0.660	0.574	0.586	-1.012	-0.945
13	2400	1.15	0.620	0.539	0.553	-1.047	-0.985
14	2600	1.15	0.600	0.522	0.522	-1.082	-1.022
15	2800	1.15	0.580	0.504	0.494	-1.117	-1.054
16	3000	1.15	0.560	0.487	0.469	-1.152	-1.083
17	3500	1.15	0.500	0.435	0.414	-1.204	-1.144
18	4000	1.15	0.446	0.388	0.370	-1.257	-1.192
19	4500	1.15	0.398	0.346	0.333	-1.274	-1.231
20	5000	1.15	0.355	0.309	0.303	-1.326	-1.263
21	5500	1.15	0.317	0.275	0.278	-1.361	-1.289
22	6000	1.15	0.283	0.246	0.256	-1.309	-1.312
23	6500	1.15	0.252	0.219	0.238	-1.379	-1.331
24	7000	1.15	0.225	0.196	0.222	-1.396	-1.347
25	7500	1.15	0.213	0.185	0.208	-1.396	-1.362
26	8000	1.15	0.201	0.175	0.195	-1.405	-1.374
27	9000	1.15	0.185	0.161	0.174	-1.414	-1.396
28	10000	1.15	0.169	0.147	0.157	-1.431	-1.413
29	11000	1.15	0.161	0.140	0.143	-1.449	-1.427
30	12000	1.15	0.145	0.126	0.131	-1.466	-1.439

Sample calculations for set 2:

- The cutoff frequency for $1\text{ k}\Omega$ resistance and $0.1\mu\text{F}$ capacitance can be calculated as

$$f = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 10^3 \times 0.1 \times 10^{-6}} = 1591.55\text{ Hz.} \quad (2.4)$$

- The time constant can be calculated as

$$\tau = R \times C = 10^3 \times 0.1 \times 10^{-6} = 0.0001\text{ s} \quad (2.5)$$

- Frequency is 200 Hz, Input voltage $V_i = 1.15\text{ V}$, Output voltage $V_o = 1.110\text{ V}$.

- Measured static gain is

$$\text{Measured gain} = \frac{V_o}{V_i} = \frac{1.110}{1.15} = 0.965 \quad (2.6)$$

- Calculated gain is estimated using the formula ($\tau = 0.0001\text{ s}$ and $\omega = 2 \times \pi \times 400 = 2513.27\text{ rad/s}$)

$$\text{Calculated gain} = \frac{1}{\sqrt{1 + \tau^2 \omega^2}} = \frac{1}{\sqrt{1 + 0.0001^2 \times 2513.27^2}} = 0.9698. \quad (2.7)$$

- The calculated phase difference can be estimated by

$$\text{Calculated phase} = \tan^{-1}(-\omega\tau) = \tan^{-1}(-2513.27 \times 0.0001) = -0.246\text{ radians} \quad (2.8)$$

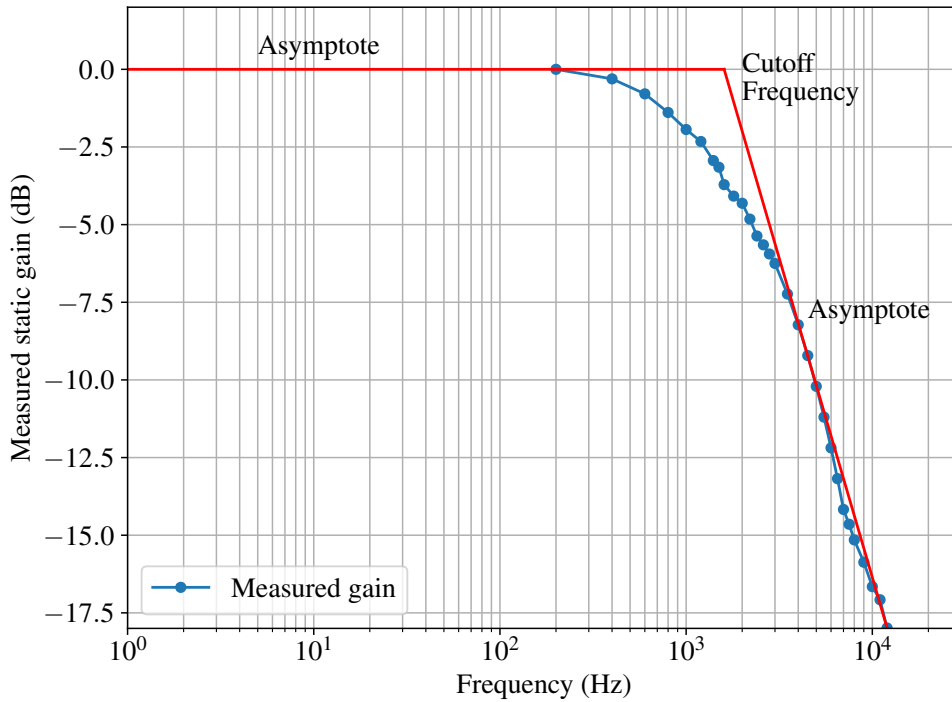


Figure 2.2: Bode plot - variation of measured static gain (in dB, Eq. 2.3) with frequency.

Figure. 2.2 is the Bode plot which shows the variation of Gain with frequency. Two asymptote is found to cut each other at cutoff frequency which is found to be 1600 Hz. Figure. 2.3 shows the comparison of measured gain with

theoretical gain. It can be seen that both theoretical and measured gain are comparable with each other. Figure. 2.4 shows the comparison between measured and theoretical phase difference which also shows no much difference.

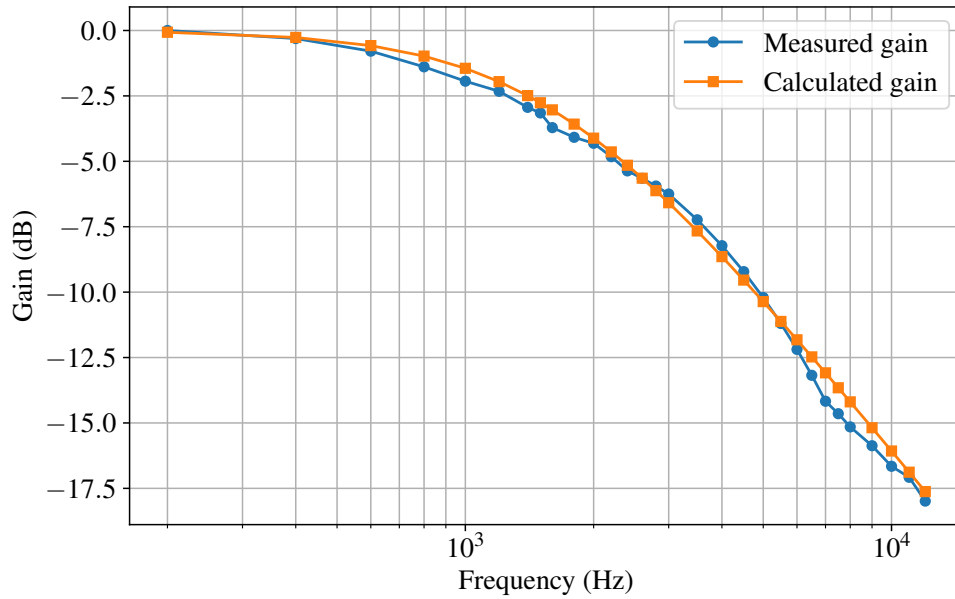


Figure 2.3: Comparison of measured and calculated gain in Decibels, varying with frequency

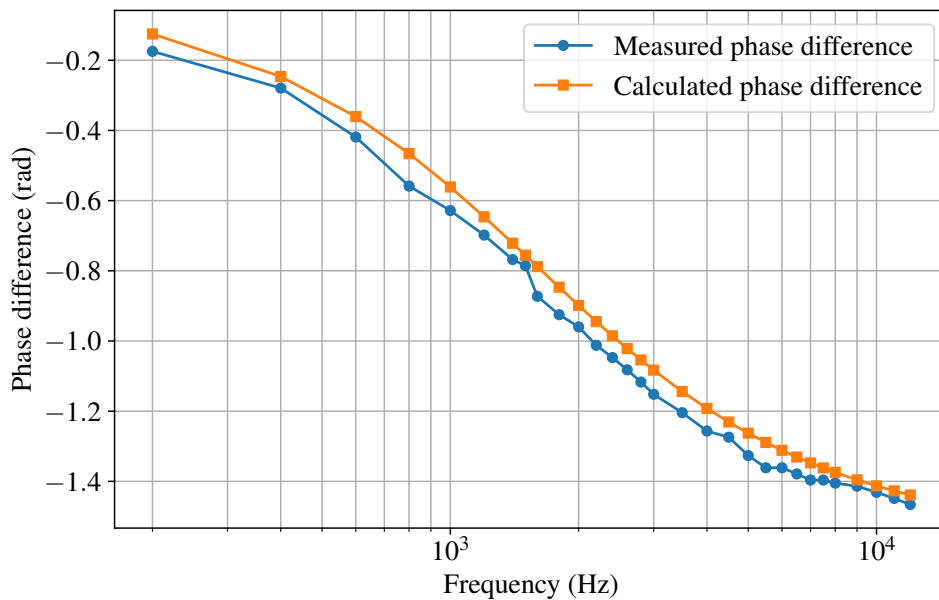


Figure 2.4: Comparison of measured and calculated phase difference in radians, varying with frequency.

The variation of output voltage with a square wave input signal is tabulated in Table. 2.2. The Fig. 2.5 shows the variation of gain with frequency for a square input signal.

Table 2.2: Variation of output voltage with frequency for square wave input

Sr. No.	Frequency (Hz)	Input peak to peak (Volts)	Measured Output Peak to peak (Volts)	Measured Output Peak to peak / Measured input peak to peak (Volts)
1	200	8	6.4	0.8
2	400	8	6.24	0.78
3	600	8	6	0.75
4	800	8	5.6	0.7
5	1000	8	4.8	0.6
6	1200	8	4	0.5
7	1400	8	3.2	0.4

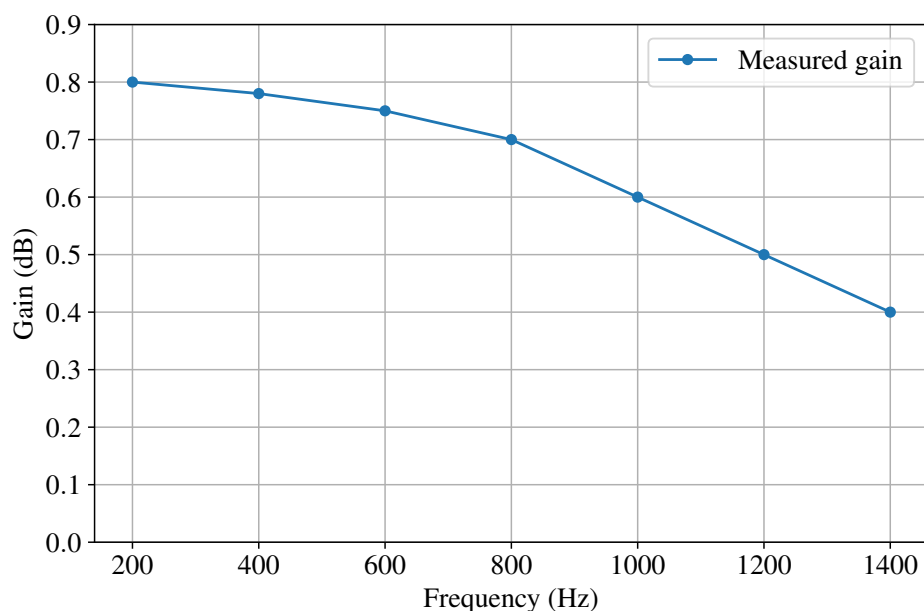


Figure 2.5: Variation of measured static gain with frequency in decibels for square wave.

Table 2.3: Sinusoidal wave input and output taken across resistor

Sl No.	Frequency (Hz)	Input peak to peak V_i (V)	Output peak to peak V_o (V)	Measured static gain (V_o/V_i)
1	400	5.0	1.196	0.239
2	800	5.0	2.13	0.426
3	1200	5.0	2.9	0.580
4	1600	5.0	3.48	0.696
5	2000	5.0	3.84	0.768
6	3000	5.0	4.36	0.872
7	4000	5.0	4.6	0.920
8	5000	5.0	4.74	0.948
9	6000	5.0	4.8	0.960

The output voltage measured across resistor for different frequencies for a sine wave input is tabulated in Ta-

ble. 2.3. The variation of measured gain with frequency is as shown in Fig. 2.6. It can be seen that the measured gain increases with increase in frequency.

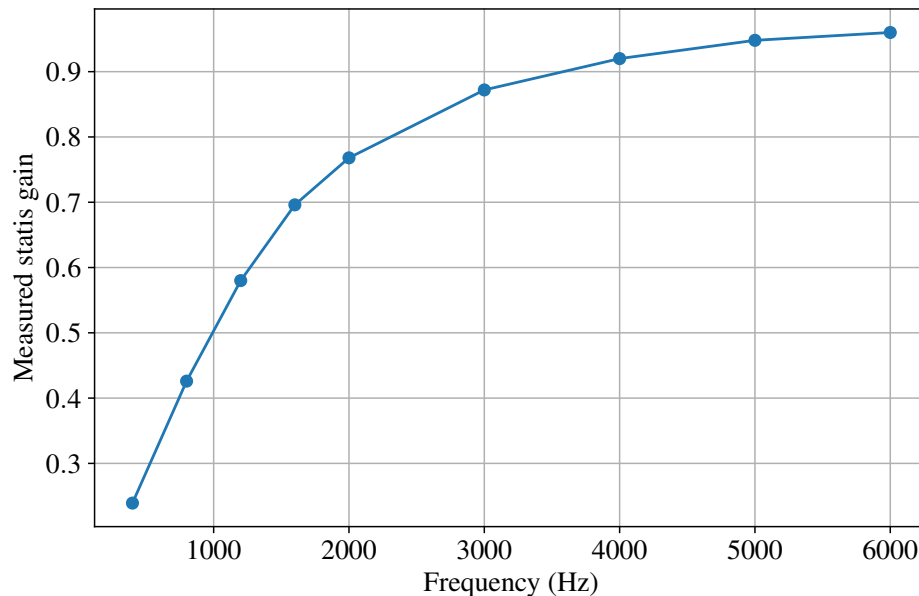


Figure 2.6: Variation of measured gain with frequency for sine wave input.

Results:

- The theoretical value of cutoff frequency for the RC circuit =1591.55 Hz
- The experimental value of cutoff frequency for the RC circuit =1600 Hz
- The time constant for RC circuit $\tau = 0.0001\text{s}$

Discussion on the Results:

- The experiment involved applying sinusoidal and square wave inputs at various frequencies to the RC circuit and analyzing the output amplitude, phase shift, and frequency response characteristics.
- The cutoff frequency for the RC circuit was calculated. This frequency is the point where the output signal starts to attenuate significantly
- The measured static gain and phase difference were recorded at various input frequencies. As expected, at frequencies below the cutoff, the circuit showed a minimal reduction in output amplitude, while at higher frequencies, the output signal amplitude decreased as the circuit acted more like a low-pass filter. This behavior was consistent with the theoretical expectations for a first-order low-pass filter.
- The Bode plot (Figure. 2.2) showed the variation of measured static gain (in dB) with frequency. The plot revealed two asymptotes, with the gain flattening out at lower frequencies and gradually decreasing above the cutoff frequency. The cutoff frequency in the experiment was observed around 1600 Hz, which closely matched the theoretical value of 1591.55 Hz.
- A comparison of measured and theoretical gain (Figure. 2.3) showed that the experimental results were in good agreement with the theoretical model, indicating that the RC circuit behaved as expected for a first-order low-pass filter.

- The phase difference between the input and output signals was also measured and compared with theoretical predictions (Figure. 2.4). The phase lag increased with frequency, as expected for a low-pass filter. At lower frequencies, the phase difference was small, while at higher frequencies, the phase difference approached $\pi/2$ radians, which is typical for a first-order system near and beyond the cutoff frequency.
- The RC circuit's response to a square wave input was also examined. The output voltage was measured at various frequencies (Table. 2.2), showing a reduction in amplitude as the frequency increased. This behavior confirmed the filtering characteristics of the RC circuit, where high-frequency components of the square wave were attenuated, leaving behind a smoother waveform at higher frequencies

Conclusions:

The experiment successfully demonstrated the dynamic response of a first-order RC circuit. The static gain decreased as the input frequency approached and exceeded the cutoff frequency, and the phase shift followed the expected trends for a first-order system. The circuit also effectively filtered a square wave input, attenuating high-frequency components and confirming its behavior as both a low-pass and high-pass filter, depending on the configuration. Both theoretical and experimental results for gain and phase difference showed excellent agreement, indicating that the RC circuit accurately simulates a first-order system. The circuit's performance as both a low-pass filter (RC Integrator) and a high-pass filter (RC Differentiator) illustrates its versatility in various filtering applications. Overall, the results validated the theoretical principles of first-order RC circuits and their use in filtering applications. This experiment reinforced the importance of RC circuits in signal processing, noise reduction, and frequency filtering

EXPERIMENT 3

FREQUENCY RESPONSE OF A SECOND ORDER SYSTEM USING RLC CIRCUIT

Aim

- To study the frequency response of a series RLC circuit for different damping ratios.
- To determine the resonance frequency in a series RLC circuit and compare its value with the theoretical value

Apparatus – Bread-board, variable resistor (potentiometer), capacitor, inductor, multi-meter, Oscilloscope and connecting wires.

Procedure:

1. Determination of resonance frequency
 - i The function generator frequency is set to 50 Hz before making the connection. The signal generator output is set to 3V (peak to peak).
 - ii Using the breadboard and wire leads connect the resistor, capacitor and inductor along with the output of the function generator to construct the circuit shown in Fig. 3.1. Note that the peak to peak voltage is measured across the resistor using the oscilloscope.
 - iii Make sure that the resistance is less than $447\ \Omega$ i.e. the system is underdamped, for Part 1 of the experiment. All the components are connected in series with the supply given by the function generator. All the ground connections are made at one end of the resistor.

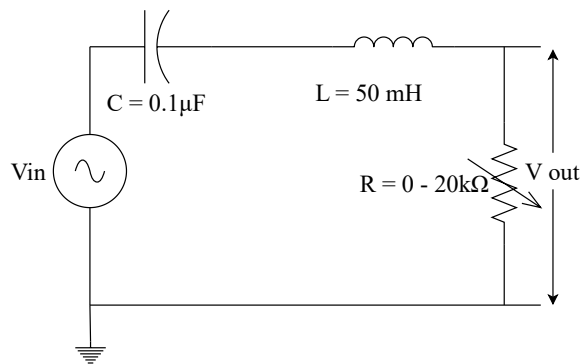


Figure 3.1: Series RLC circuit for resonance frequency

- iv Use following relation to compute the expected resonance frequency and record your results in Table 3.1

$$\omega_n = \frac{1}{\sqrt{LC}}$$

- v Set the function generator frequency to 100 Hz and record the peak to peak voltage from the oscilloscope. Then, adjust the excitation frequency to 200 Hz and record the voltage. Adjust the frequency to 300 Hz and record the voltage. Continue varying the excitation frequency to discrete values below the expected resonance frequency computed in step (iv). Record the voltage for each of these values.
- vi Determine the experimental value for resonance frequency by finding the frequency that produces the largest voltage on the oscilloscope. Record this frequency and voltage.
- vii Record the voltage for frequency values that are above the resonance frequency determined in step (iv).

viii Plot a graph between voltage and frequency (f) and determine the resonance frequency of the circuit.

2. Frequency response of RLC circuit for different damping ratios

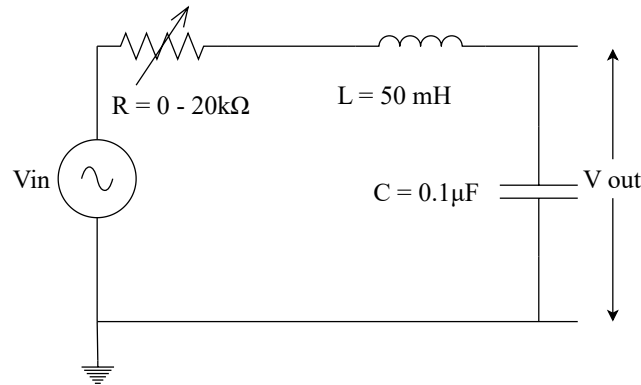


Figure 3.2: Series RLC circuit for different damping ratios

- i Using the breadboard and wire leads connect the resistor, capacitor and inductor along with the output of the function generator to construct the circuit shown in Figure 3.2. Note that the peak to peak voltage is measured across the capacitor using the oscilloscope.
- ii The function generator frequency is set to 100 Hz before making the connection. The signal generator output is set to 3V (peak to peak).
- iii Use the same circuit as used in Part (1) of this experiment. In this case, the output voltage is measured across the capacitor, so connect the terminals of the oscilloscope across the capacitor.
- iv The damping ratio ζ is given by

$$\zeta = \frac{R}{2} \sqrt{\frac{C}{L}} \quad (3.9)$$

Vary the value of the resistance such that L takes different values, so that overdamping, underdamping and critical damping in the circuit is observed

- v Adjust the resistance such that the system corresponds to an underdamped system. Apply a frequency of 100 Hz and record the input and output voltage. Change the excitation frequency as integer multiples of resonance frequency ($\omega = \omega_n, \omega_{2n}, \omega_{3n}, \omega_{4n}$) and obtain the input and output voltages.
- vi Repeat step v for different damping ratios. Take 3 sets of readings. One reading for under-damped and over-damped system, and one reading for critically damped case. (You can change the damping ratio by changing the resistance value of the variable resistance).
- vii Plot a graph between V_o/V_i and ω/ω_n for different values of L .
- viii Compare the experimental and theoretical behavior of the system. Theoretically, the ratio V_o/V_i of a second order system is given as

$$\left| \frac{V_o}{V_i} \right| = \frac{1}{\sqrt{(1 - \frac{\omega^2}{\omega_n^2})^2 + (\frac{2\zeta\omega}{\omega_n})^2}} \quad (3.10)$$

Observations :

Inductance = 50 mH
 Capacitance = 0.1 μ F
 Resistance = 1-20 k Ω

- The Table. 3.1 shows the output voltage of an LCR circuit, recorded for a constant input voltage at different frequency for a damping ratio of $\zeta = 0.1$. The damping ratio $\zeta = 0.1$ is achieved by calculating the resistance value as per Eq. 3.9 given by

$$R = \zeta \times 2 \times \sqrt{\frac{L}{C}} = 0.1 \times 2 \times \sqrt{\frac{50 \times 10^{-3}}{0.1 \times 10^{-6}}} = 141.42 \Omega \quad (3.11)$$

- The theoretical resonance frequency is calculated using the formula

$$\omega_n = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{50 \times 10^{-3} \times 0.1 \times 10^{-6}}} = 14142.14 \text{ rad/s} = 2250.8 \text{ Hz (} f_n \text{)} \quad (3.12)$$

- The variation of output voltage with frequency for a constant input voltage with a damping ratio of $\zeta = 0.1$ is tabulated in Table. 3.1.

Table 3.1: Variation of Output Voltage with Frequency for $\zeta = 0.1$

S/ No.	Frequency f (Hz)	Input Voltage V_i (V)	Output Voltage V_o (V)
1	200	6.5	6.5
2	400	6.5	6.7
3	600	6.5	6.8
4	800	6.5	7
5	1000	6.5	7.6
6	1200	6.5	8
7	1400	6.5	8.5
8	1600	6.5	9.2
9	1800	6.5	10.7
10	2000	6.5	11.9
11	2200	6.5	13.3
12	2400	6.5	15
13	2600	6.5	17
14	2700	6.5	16.9
15	2800	6.5	16.3
16	3000	6.5	14
17	3200	6.5	12
18	3400	6.5	9
19	3600	6.5	7.2
20	3800	6.5	6
21	4000	6.5	5.2
22	4500	6.5	4
23	5000	6.5	3.4
24	5500	6.5	2.82
25	6000	6.5	2.6

- The variation of output voltage with frequency is plotted and shown in Fig. 3.3. From the plot, the resonant frequency of the system is found to be $f_{expt} = 2600 \text{ Hz}$.
- The percentage difference between the experimental and theoretical value of the resonant frequency is calculated using

$$\text{Deviation} = \frac{f_n - f_{expt}}{f_n} \times 100 = 15.5 \% \quad (3.13)$$

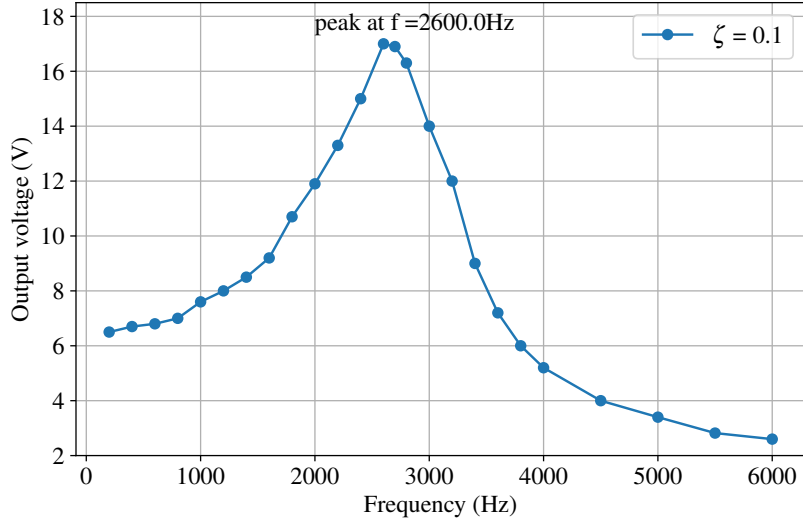


Figure 3.3: Variation of output voltage with frequency

- The Table. 3.2 shows the output voltage of an LCR circuit, recorded for a constant input voltage at different frequency for a damping ratio of $\zeta = 0.2$ and its deviation from the theoretical output voltage. The damping ratio $\zeta = 0.2$ is achieved by calculating the resistance value as per Eq. 3.9 given by

$$R = \zeta \times 2 \times \sqrt{\frac{L}{C}} = 0.2 \times 2 \times \sqrt{\frac{50 \times 10^{-3}}{0.1 \times 10^{-6}}} = 282.84 \Omega \quad (3.14)$$

- The Table. 3.3 shows the output voltage of an LCR circuit, recorded for a constant input voltage at different frequency for a damping ratio of $\zeta = 1$ and its deviation from the theoretical output voltage. The damping ratio $\zeta = 1$ is achieved by calculating the resistance value as per Eq. 3.9 given by

$$R = \zeta \times 2 \times \sqrt{\frac{L}{C}} = 1 \times 2 \times \sqrt{\frac{50 \times 10^{-3}}{0.1 \times 10^{-6}}} = 1414.2 \Omega \quad (3.15)$$

- The Table. 3.4 shows the output voltage of an LCR circuit, recorded for a constant input voltage at different frequency for a damping ratio of $\zeta = 2$ and its deviation from the theoretical output voltage. The damping ratio $\zeta = 0.2$ is achieved by calculating the resistance value as per Eq. 3.9 given by

$$R = \zeta \times 2 \times \sqrt{\frac{L}{C}} = 2 \times 2 \times \sqrt{\frac{50 \times 10^{-3}}{0.1 \times 10^{-6}}} = 2828.4 \Omega \quad (3.16)$$

The variation of V_o/V_i with ω/ω_n is plotted for the damping ratios $\zeta = 0.2$, $\zeta = 1$ and $\zeta = 2$ and is shown in Fig. 3.4. From the figure, it can be seen that for underdamped system, the ratio of voltages increases and reaches peak at resonance and then decreases with increase in frequency. For the case of critical damping and overdamping, the voltage ratio decreases with increase in frequency ratio. The theoretical voltage ratio is also plotted with theoretical frequency ratio which is shown in Fig. 3.5.

Table 3.2: Variation of output voltage with frequency for an under damped system for a damping ratio of $\zeta = 0.2$

Sl No.	Frequency (Hz)	Voltage input V_i (V)	Measured output V_o (V)	Theoretical output V_o (V)	Deviation (%)
1	200	6.5	6.5	6.536	0.544
2	400	6.5	6.7	6.644	0.837
3	600	6.5	6.8	6.833	0.486
4	800	6.5	7.0	7.114	1.607
5	1000	6.5	7.5	7.507	0.094
6	1200	6.5	7.8	8.041	2.998
7	1400	6.5	8.3	8.760	5.251
8	1600	6.5	8.8	9.726	9.524
9	1800	6.5	9.5	11.021	13.803
10	2000	6.5	10.1	12.714	20.561
11	2200	6.5	10.9	14.711	25.906
12	2400	6.5	12.0	16.341	26.567
13	2600	6.5	12.5	16.250	23.077
14	2700	6.5	12.3	15.377	19.358
15	2800	6.5	11.5	14.148	18.714
16	3000	6.5	10.1	11.440	11.715
17	3200	6.5	8.3	9.125	9.044
18	3400	6.5	6.9	7.370	6.380
19	3600	6.5	5.7	6.067	6.045
20	3800	6.5	4.9	5.087	3.682
21	4000	6.5	4.3	4.336	0.835
22	4500	6.5	3.2	3.077	3.987
23	5000	6.5	2.6	2.317	12.229
24	5500	6.5	2.2	1.817	21.047
25	6000	6.5	1.5	1.4770	2.066

Sample calculation for set 2:

- Voltage input $V_i = 6.5$ V, Measured voltage output $V_o = 6.7$ V.
- The theoretical voltage is calculated using Eq. 3.10 given by

$$\left| \frac{V_o}{V_i} \right| = \frac{1}{\sqrt{(1 - \frac{\omega^2}{\omega_n^2})^2 + (\frac{2\zeta\omega}{\omega_n})^2}} \quad (3.17)$$

where $\omega = 2513.27$ rad/s, $\omega_n = 16336.28$ rad/s, $\zeta = 0.2$. Substituting these values gives

$$\text{Theoretical output voltage} = \frac{1}{\sqrt{(1 - \frac{2513.27^2}{16336.28^2})^2 + (\frac{2 \times 0.2 \times 2513.27}{16336.28})^2}} = 6.644 \text{ V} \quad (3.18)$$

- Deviation is calculated by

$$\text{Deviation} = \frac{\text{Theoretical output voltage} - \text{Measured output voltage}}{\text{Theoretical output voltage}} \times 100 = 0.837\% \quad (3.19)$$

Table 3.3: Variation of output voltage with frequency for a critically damped system for a damping ratio of $\zeta = 1$

Sl No.	Frequency (Hz)	Voltage input V_i (mV)	Measured output V_o (mV)	Theoretical output V_o (mV)	Deviation (%)
1	200	6.5	6.5	6.642	0.592
2	400	6.5	6.45	6.350	1.579
3	600	6.5	6.3	6.171	2.085
4	800	6.5	6.0	5.938	1.047
5	1000	6.5	5.9	5.662	4.197
6	1200	6.5	5.5	5.359	2.640
7	1400	6.5	5.3	5.039	5.180
8	1600	6.5	5.0	4.715	6.054
9	1800	6.5	4.7	4.394	6.964
10	2000	6.5	4.3	4.084	5.298
11	2200	6.5	4.1	3.788	8.239
12	2400	6.5	3.8	3.510	8.275
13	2600	6.5	3.5	3.250	7.692
14	2700	6.5	3.4	3.127	8.716
15	2800	6.5	3.3	3.010	9.650
16	3000	6.5	3.1	2.788	11.188
17	3200	6.5	2.9	2.585	12.198
18	3400	6.5	2.7	2.398	15.572
19	3600	6.5	2.4	2.228	7.711
20	3800	6.5	2.2	2.073	6.145
21	4000	6.5	2.0	1.931	3.596
22	4500	6.5	1.6	1.627	1.648
23	5000	6.5	1.4	1.384	1.193
24	5500	6.5	1.3	1.187	9.497
25	6000	6.5	1.25	1.028	21.643

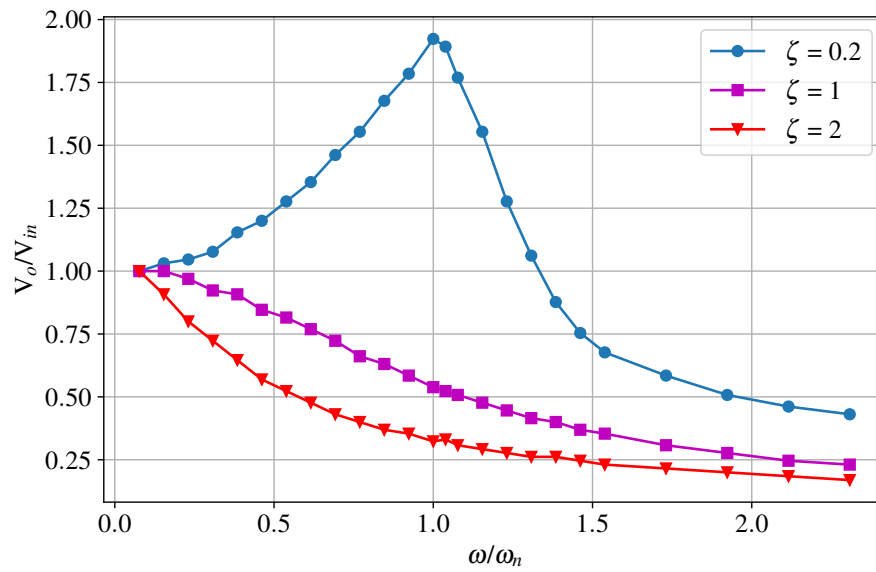


Figure 3.4: Comparison of variation of V_o/V_i with ω/ω_n for different damping ratios.

Table 3.4: Variation of output voltage with frequency for an over damped system for a damping ratio of $\zeta = 2$

Sl No.	Frequency (Hz)	Voltage input V_i (mV)	Measured output V_o (mV)	Theoretical output V_o (mV)	Deviation (%)
1	200	6.5	6.5	6.246	4.061
2	400	6.5	5.9	5.632	4.756
3	600	6.5	5.2	4.916	5.782
4	800	6.5	4.7	4.254	10.477
5	1000	6.5	4.2	3.696	13.637
6	1200	6.5	3.7	3.239	14.239
7	1400	6.5	3.4	2.866	18.627
8	1600	6.5	3.1	2.560	21.078
9	1800	6.5	2.8	2.307	21.380
10	2000	6.5	2.6	2.094	24.156
11	2200	6.5	2.4	1.914	25.410
12	2400	6.5	2.2	1.759	25.071
13	2600	6.5	2.0	1.625	23.077
14	2700	6.5	1.95	1.565	24.638
15	2800	6.5	1.85	1.508	22.688
16	3000	6.5	1.7	1.405	21.021
17	3200	6.5	1.55	1.313	18.037
18	3400	6.5	1.4	1.231	13.696
19	3600	6.5	1.3	1.158	12.278
20	3800	6.5	1.2	1.091	9.948
21	4000	6.5	1.1	1.031	6.680
22	4500	6.5	0.95	0.902	5.303
23	5000	6.5	0.85	0.797	6.601
24	5500	6.5	0.7	0.711	1.491
25	6000	6.5	0.6	0.638	5.902

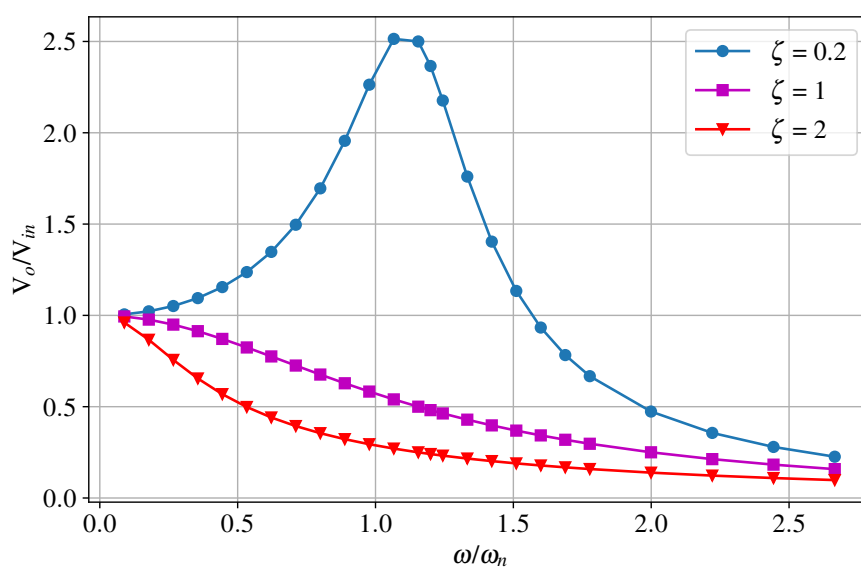


Figure 3.5: Comparison of variation of theoretical V_o/V_i with ω/ω_n for different damping ratios.

Results:

- The frequency response of the series RLC circuit was studied under various damping conditions
- Theoretical resonance frequency = 2250.8 Hz
- Experimental resonance frequency = 2600 Hz.
- Percentage deviation = 15.5%

Discussion on the Results:

- The experiment aimed to study the frequency response of a series RLC circuit under different damping conditions, and to determine the resonance frequency experimentally while comparing it with the theoretical value. The key objectives were achieved by constructing the circuit on a breadboard, varying the resistance to observe the effect of different damping ratios, and plotting the amplitude ratio versus the frequency ratio.
- The theoretical resonance frequency is equal to 2250.8 Hz. The experimentally observed resonance frequency, however, was 2600 Hz, deviating by about 15.5% from the theoretical prediction. This deviation can be attributed to factors such as: Component tolerances (resistor, capacitor, and inductor) that could vary from their nominal values. Measurement errors, including inaccuracies from the oscilloscope or the function generator. Parasitic inductance and capacitance from the breadboard and wiring, which could shift the resonance point slightly.
- The experimental data showed a typical resonance curve, where the output voltage initially increases with frequency, reaches a peak at the resonance frequency, and then decreases as the frequency continues to increase. The voltage peak corresponds to the resonance condition, where the impedance of the RLC circuit is minimized, and maximum energy is transferred from the input signal to the circuit. This was observed clearly in the graph of output voltage versus frequency (Figure. 3.3), where the highest voltage was recorded at 2600 Hz.
- The frequency response was also analyzed for different damping ratios ($\zeta = 0.1, 1$ and 2), which are crucial in determining the behavior of the circuit near resonance.
 - Underdamped case ($\zeta = 0.1$): For the underdamped system, the output voltage reaches a sharp peak at resonance and then decays as the frequency increases beyond resonance. This is characteristic of an under damped system, where there is a large oscillatory response.
 - Critically damped case ($\zeta = 1$): In the critically damped case, the voltage response shows a much smoother curve, with no sharp resonance peak. The system does not oscillate, and the response decays more steadily after resonance.
 - Over damped case ($\zeta = 2$): For the over damped system, the voltage response is significantly smaller and decays rapidly as the frequency increases. There is no pronounced peak, and the system is slow to respond to changes in frequency.
- These observations align with theoretical predictions, where the damping ratio affects the sharpness and height of the resonance peak. As the damping ratio increases, the system becomes less responsive at resonance, and the peak of the voltage response diminishes.
- The measured values were in close agreement with the theoretical values, with some deviations. The deviations were especially noticeable in the critically damped and over damped systems, where the measured output was generally lower than the theoretical values. These discrepancies can be attributed to the non-ideal nature of the components and potential inaccuracies in the damping ratio calculation due to component tolerances.

Conclusions:

In this experiment, we successfully studied the frequency response of a series RLC circuit under different damping conditions. The resonance frequency was determined experimentally and compared with theoretical calculations, showing a small but significant deviation of about 15.5%. The frequency response curves demonstrated the expected behavior for under damped, critically damped, and over damped systems, with the resonance frequency serving as the key point where the output voltage was maximal in the under damped case. The experimental results also revealed that the voltage response became smoother and smaller with increasing damping, as predicted theoretically. Resonance and damping are essential in applications like mechanical systems (vehicle suspensions), electrical circuits (tuning and filtering), control systems (stabilizing responses), and signal processing (selecting frequencies), where proper damping ensures optimal performance, stability, and efficiency. By adjusting damping, engineers can fine-tune system responses to avoid oscillations, improve stability, and enhance frequency selectivity. Despite minor discrepancies between experimental and theoretical results, the experiment successfully illustrated the principles of resonance and damping in a second-order system. The deviations could be reduced with more precise component selection and measurements.

EXPERIMENT 4

PART A: APPLICATION OF OP-AMP AS NON-INVERTING AMPLIFIER

Aim - To set-up the circuit of op-amp as non-inverting amplifier and measure its gain.

Apparatus – IC 741, Resistors – 1 k Ω , 100 Ω , 10 k Ω , Regulated DC power supply, Oscilloscope, Multimeter.

Circuit Diagram

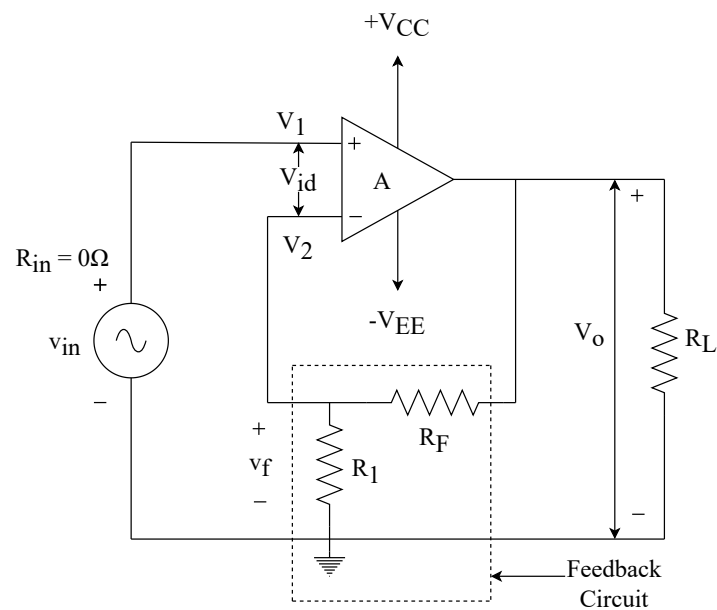


Figure 4.1: Circuit of Non-inverting amplifier

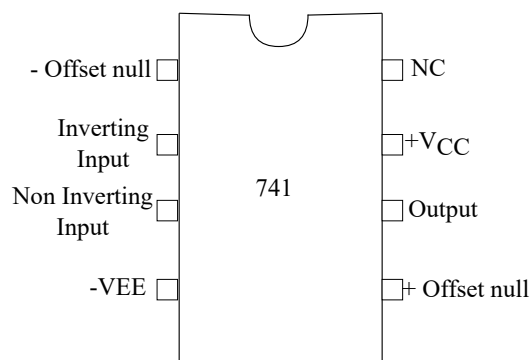


Figure 4.2: Schematic of IC741 (Op-AMP)

Design Procedure:

- Select the desired gain of the amplifier.
- The ideal gain of non-inverting amplifier is given by

$$A_F = 1 + \frac{R_f}{R_i} \quad (4.20)$$

- For Ex: Gain = 11; $R_1 = 1 \text{ k}\Omega$ $R_F = 10 \text{ k}\Omega$ OR $R_1 = 100 \text{ }\Omega$ $R_F = 1 \text{ k}\Omega$ (Gain is dependent on the ratio of resistors, not the absolute values)
- The exact gain of non-inverting amplifier is given by

$$A_F = \frac{V_o}{V_i} = \frac{A(R_1 + R_f)}{R_1 + R_f + AR_1} \quad (4.21)$$

Experimental Procedure:

- Connect the circuit as shown in Fig. 4.1.
- Apply +12 V to pin 7 and –12 V to pin 4. Connect common terminal of power supply to ground on the breadboard.
- Apply an A.C voltage of 0.2 V using Oscilloscope to the pin-3 of IC. Measure the output using oscilloscope.
- Increase voltage in steps of 0.2 V up to 1.0 V and measure the output. Note down the output in observation table.
- Change the value of all resistance such that the signal after amplification does not exceed 10 V.
- Apply a D.C voltage of 1 V using Oscilloscope to the pin-3 of IC. Measure the output.
- Increase voltage in steps of 0.5 V up to 2.0 V and measure the output. Note down the output in observation table.

Obeservations:

Feed back resistor= $R_f = 10\text{k}\Omega$, Input resistor = $R_1 = 1\text{k}\Omega$.

Open loop gain $A = 10^6$.

Table 4.1: Non-inverting Amplifier gain

S. No.	Type of power supply to input	Non-Inverting input voltage (V)	Output voltage (V)	Measured Gain	Ideal Gain	Deviation (%)	Exact Gain	Deviation (%)
1	A.C	0.208	2.24	10.769	11	2.098	10.99	2.009
2	A.C	0.408	4.48	10.980	11	0.178	10.99	0.087
3	A.C	0.608	6.6	10.855	11	1.316	10.99	1.226
4	A.C	0.84	9	10.714	11	2.597	10.99	2.509
5	A.C	1	10.9	10.900	11	0.909	10.99	0.819
6	D.C	0.2	2.1	10.500	11	4.545	10.99	4.459
7	D.C	0.4	4.38	10.950	11	0.455	10.99	0.364
8	D.C	0.6	6.5	10.833	11	1.515	10.99	1.426
9	D.C	0.8	8.6	10.750	11	2.273	10.99	2.184
10	D.C	1	10.9	10.900	11	0.909	10.99	0.819

Sample calculations for set 2:

- Non inverting voltage input $V_i = 0.408$ V, Output voltage $V_o = 4.48$ V
- Measured gain is calculated using

$$\text{Measured gain} = \frac{V_o}{V_i} = \frac{4.48}{0.408} = 10.980 \quad (4.22)$$

- Ideal gain can be calculated using Eq. 4.20 as

$$\text{Ideal gain} = 1 + \frac{R_f}{R_1} = 1 + \frac{10}{1} = 11 \quad (4.23)$$

- Deviation from calculated gain is calculated as

$$\text{Deviation} = \frac{\text{Ideal gain} - \text{Measured gain}}{\text{Ideal gain}} = \frac{11 - 10.980}{11} \times 100 = 0.178 \% \quad (4.24)$$

- Exact gain can be calculated using Eq. 4.21 as

$$\text{Exact gain} = \frac{A(R_1 + R_f)}{R_1 + R_f + AR_1} = \frac{10^6(1 + 10)}{1 + 10 + 10^6 * 1} = 10.99 \quad (4.25)$$

- Deviation from exact gain is calculated as

$$\text{Deviation} = \frac{\text{Exact gain} - \text{Measured gain}}{\text{Exact gain}} = \frac{10.99 - 10.98}{10.99} \times 100 = 0.087 \% \quad (4.26)$$

Results and discussions:

- In this experiment, an op-amp was configured as a non-inverting amplifier, and its gain was measured for different input voltages (both AC and DC). The measured gain for the AC input ranged from 10.769 to 10.9, and for the DC input, it ranged from 10.5 to 10.9. The deviation from the ideal gain for both AC and DC inputs was generally very small, with a maximum deviation of approximately 2.6% for AC and 4.5% for DC. The exact gain, calculated using the op-amp's open-loop gain $A=10^6$, was close to 10.99 for all measurements, with deviations less than 1% from the measured value.
- The experiment successfully demonstrated the operation of an operational amplifier (op-amp) configured as a non-inverting amplifier, allowing for the accurate measurement of its gain.
- The measured gain was consistently positive, with the output voltage signal remaining in phase with the input signal, which aligns with theoretical expectations for this configuration.
- In the non-inverting setup, the input voltage signal (V_{in}) is applied directly to the non-inverting terminal, resulting in a positive gain that preserves the phase relationship between input and output.
- Negative feedback is established by feeding a portion of the output voltage back to the inverting terminal via a resistor divider network (R_f and R_1), enhancing the amplifier's stability and input impedance.
- The high input impedance is advantageous as it minimizes the loading effect on the input signal source, preserving the original characteristics of the signal.
- The closed-loop gain is always greater than or equal to one, with the gain determined by the ratio of R_f to R_1 . Adjusting these resistors allows for precise control over the gain.

- The LM741 op-amp was chosen for its versatility, reliability, and suitability for educational purposes, making it ideal for this experiment. Its straightforward pin configuration and performance characteristics support various applications.
- It is essential to consider that in certain configurations (like differentiators or integrators), the expected phase shifts (e.g., 90 degrees) can vary due to factors such as component tolerances, loading effects, and frequency response.
- Deviations from expected phase shifts could indicate issues, including incorrect component values, unexpected loading conditions, or parasitic effects in the circuit.

Conclusions:

The experiment successfully demonstrated the operation of an op-amp configured as a non-inverting amplifier. The measured gain closely matched the theoretical and exact gain values, with minimal deviation. This confirms the proper functioning of the non-inverting amplifier configuration. The small deviations observed are likely due to practical limitations, such as component tolerances and the finite open-loop gain of the op-amp. The feedback circuit is used to stabilize and control the amplifier's gain, ensuring linear operation, improving input impedance, and enhancing overall performance by reducing the impact of component variations. This experiment highlights the importance of feedback in controlling the gain of op-amp circuits and demonstrates how the circuit can be used in various applications such as signal conditioning, audio amplification, and sensor interfacing, where a stable, predictable gain is required.

EXPERIMENT 4

PART B: APPLICATION OF OP-AMP AS INVERTING AMPLIFIER

Aim – To set-up the circuit of op-amp as inverting amplifier and measure its gain

Apparatus – IC 741, Resistors – 1 kΩ, 100 Ω, 10 kΩ, Regulated DC powersupply, Oscilloscope, Multimeter.

Circuit Diagram

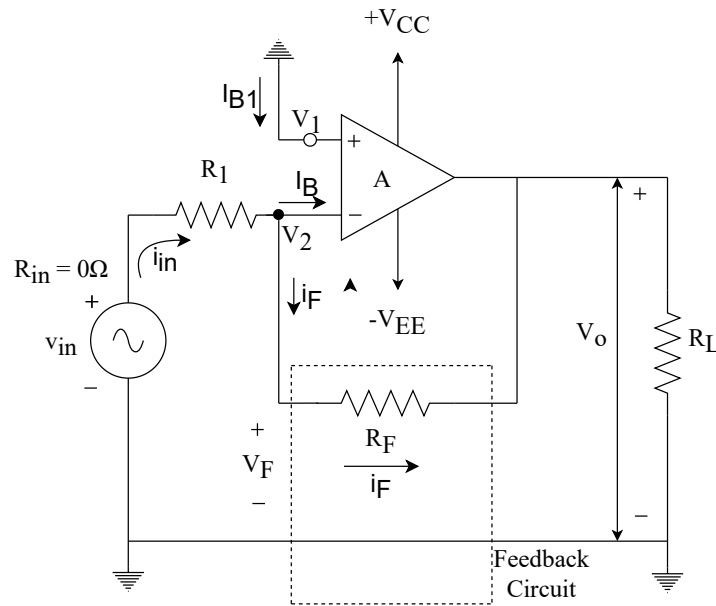


Figure 4.3: Circuit of inverting amplifier

Design Procedure

- Select the desired gain of the amplifier.
- The ideal gain of non-inverting amplifier is given by

$$A_F = \frac{V_o}{V_i} = \frac{AR_F}{AR_i} = \frac{R_F}{R_i} \quad (4.27)$$

- For Ex: Gain = -10; $R_1 = 1 \text{ k}\Omega$ $R_F = 10 \text{ k}\Omega$ OR $R_1 = 100 \text{ }\Omega$ $R_F = 1 \text{ k}\Omega$ (Gain is dependent on the ratio of resistors, not the absolute values)
- The exact gain of non-inverting amplifier is given by

$$\frac{V_o}{V_i} = -\frac{AR_F}{R_1 + R_F + AR_i} \quad (4.28)$$

Experimental Procedure

- Connect the circuit as shown in Fig. 4.3
- Apply +12 V to pin 7 and -12 V to pin 4. Connect common terminal of power supply to ground on the breadboard.

- Apply an A.C voltage of 0.2 V using PC Oscilloscope to the pin-2 of IC. Measure the output.
- Increase voltage in steps of 0.2 V up to 1.0 V and measure the output. Note down the output in observation table.
- Change the value of all resistance such that the signal after amplification does not exceed 10 V.
- Apply a D.C voltage of 1 V using Oscilloscope to the pin-2 of IC. Measure the output.
- Increase voltage in steps of 0.5 V up to 2.0 V and measure the output. Note down the output in observation table.

Observations:

Input resistance = $R_1 = 1k\Omega$, Feed back Resistance = $R_f = 10k\Omega$.

Open loop gain $A = 10^6$

Table 4.2: Inverting Amplifier Gain

Sr. No.	Type of Input	Inverting input voltage (V)	Output voltage (V)	Measured Gain	Ideal Gain	Deviation (%)	Exact Gain	Deviation (%)
1	A.C	0.2	-2.1	-10.50	-10	5	-9.99	5.01
2	A.C	0.4	-4.1	-10.25	-10	2.5	-9.99	2.51
3	A.C	0.6	-6.0	-10.00	-10	0.0	-9.99	0.01
4	A.C	0.8	-7.72	-9.65	-10	3.5	-9.99	3.49
5	A.C	1.0	-10.0	-10.00	-10	0.0	-9.99	0.01
6	D.C	0.2	-1.96	-9.80	-10	2.0	-9.99	1.99
7	D.C	0.4	-3.86	-9.65	-10	3.5	-9.99	3.49
8	D.C	0.6	-5.84	-9.73	-10	2.67	-9.99	2.66
9	D.C	0.8	-7.76	-9.70	-10	3.0	-9.99	2.99
10	D.C	1.0	-9.68	-9.68	-10	3.2	-9.99	3.19

Sample calculations for set 2:

- Non inverting voltage input $V_i = 0.4$ V, Output voltage $V_o = -4.1$ V
- Measured gain is calculated using

$$\text{Measured gain} = \frac{V_o}{V_i} = \frac{-4.1}{0.4} = -10.25 \quad (4.29)$$

- Ideal gain can be calculated using Eq. 4.27 as

$$\text{Ideal gain} = -\frac{R_f}{R_1} = -\frac{10}{1} = -10 \quad (4.30)$$

- Deviation from calculated gain is calculated as

$$\text{Deviation} = \frac{\text{Ideal gain} - \text{Measured gain}}{\text{Ideal gain}} = \left| \frac{-10 + 10.25}{-10} \right| \times 100 = 2.5 \% \quad (4.31)$$

- Exact gain can be calculated using Eq. 4.21 as

$$\text{Exact gain} = -\frac{AR_F}{R_1 + R_F + AR_1} = \frac{10^6 \times 10}{1 + 10 + 10^6 \times 1} = -9.99 \quad (4.32)$$

- Deviation from exact gain is calculated as

$$\text{Deviation} = \frac{\text{Exact gain} - \text{Measured gain}}{\text{Exact gain}} = \left| \frac{-9.99 + 10.25}{-9.99} \right| \times 100 = 2.51 \% \quad (4.33)$$

Results and discussions:

- In this experiment, an op-amp was configured as a non-inverting amplifier, and its gain was measured for different input voltages (both AC and DC). The measured gain for the AC input ranged from 10.769 to 10.9, and for the DC input, it ranged from 10.5 to 10.9. The deviation from the ideal gain for both AC and DC inputs was generally very small, with a maximum deviation of approximately 2.6% for AC and 4.5% for DC. The exact gain, calculated using the op-amp's open-loop gain $A=10^6$, was close to 10.99 for all measurements, with deviations less than 1% from the measured value.
- The primary focus was to measure the gain of the circuit and compare the experimentally obtained values with the ideal and exact theoretical values. It was observed that the gain of the inverting amplifier was negative, indicating that the output signal is inverted and out of phase with the input signal.
- In the inverting amplifier configuration, the input voltage is applied to the inverting (–) terminal of the op-amp, while the non-inverting (+) terminal is typically grounded. This setup produces a phase shift of 180 degrees between the input and output signals, meaning that when the input signal increases, the output signal decreases correspondingly. This negative gain characteristic is a fundamental feature of inverting amplifiers
- The findings of this experiment highlight the essential characteristics of the inverting operational amplifier configuration, particularly its negative gain and the resulting phase inversion of the output signal. These features enable a wide range of practical applications.

Conclusions:

The experiment successfully illustrated the operation of an inverting amplifier using an op-amp. The results were consistent with theoretical predictions, with small deviations attributed to real-world component tolerances and parasitic effects. The feedback mechanism was essential in achieving the desired gain and stabilizing the amplifier's performance. This study of inverting amplifiers is crucial for applications where signal inversion and amplification are required, such as in audio processing, signal conditioning, and instrumentation systems. . Additionally, inverting amplifiers are integral to active filtering applications, allowing for precise control over circuit frequency response. Their capability to perform mathematical operations makes them essential in analog computing systems, while their function in feedback control systems ensures effective regulation of output in response to varying inputs.

EXPERIMENT 4

PART C: APPLICATION OF OP-AMP AS DIFFERENTIAL AMPLIFIER

Aim – To set-up the circuit of op-amp as differential amplifier and measure its gain

Apparatus – IC 741, Resistors – 1 k Ω , 100 Ω , 10 k Ω , Regulated DC power supply, Multimeter.

Circuit Diagram

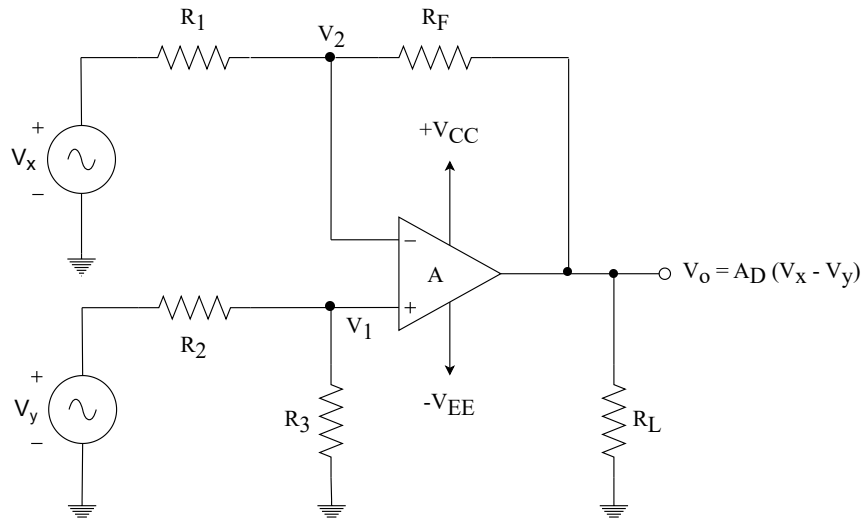


Figure 4.4: Circuit of differential amplifier

Design Procedure:

- Select the desired gain of the amplifier.
- The ideal gain of the differential amplifier is given by

$$A_D = -\frac{V_o}{V_x - V_y} = -\frac{R_F}{R_1} \quad (4.34)$$

- Use $R_2 = R_1$ and $R_F = R_3$

Experimental Procedure:

- Connect the circuit as shown in Fig. 4.4.
- Apply +12 V to pin 7 and –12 V to pin 4. Connect common terminal of power supply to ground on the breadboard.
- Apply two different voltages V_x and V_y and measure the output. Make sure that the difference in the voltage should not exceed 10 V.
- Repeat the above step for different V_x and V_y

Observations:

Input resistance $R_1 = R_2 = 1\text{k}\Omega$, Feedback Resistance $R_3 = R_f = 10\text{k}\Omega$

Table 4.3: Differential Amplifier Gain

Sr No.	Input voltage (V_x) (Volts)	Input voltage (V_y) (Volts)	Measured Output voltage (V_o) (Volts)	Theoretical output voltage	Deviation (%)
1	0.4	2	15.5	16	3.75
2	0.8	2	11.8	12	1.66
3	1.2	2	7.8	8	2.55
4	1.6	2	4.2	4	5.0
5	2.0	2	0	0	0

Sample calculations for set 2:

- Input voltage $V_x = 0.8$ V, $V_y = 2$ V and Measured output voltage $V_o = 11.8$ V
- Gain can be calculated using Eq. 4.34 as

$$\text{Ideal gain } A_D = -\frac{R_F}{R_1} = -\frac{10}{1} = -10 \quad (4.35)$$

- Theoretical output voltage also can be calculated using the Eq. 4.34 as

$$\text{Theoretical voltage output} = A_D(V_x - V_y) = -10(0.8 - 2) = 12 \text{ V} \quad (4.36)$$

- Deviation of measured voltage output and theoretical voltage output is calculated as

$$\text{Deviation} = \frac{\text{Theoretical output} - \text{Measured output}}{\text{Measured output}} = \frac{12 - 11.8}{12} \times 100 = 1.66\% \quad (4.37)$$

Results and discussions:

- The op-amp was configured as a differential amplifier in this experiment with two input voltages V_x and V_y . For each pair of input voltages, the output voltage V_o was measured, and the theoretical output voltage was calculated.
- The measured output voltage for each input combination was compared with the theoretical output voltage. The deviations between the measured and theoretical output voltages were recorded, with deviations ranging from 0% to 5%.
- The operation of an operational amplifier (Op-amp) functioning as a differential amplifier was thoroughly examined.
- During this analysis, it was observed that the gain of the amplifier was negative. From the data, it is evident that as the input voltage V_x increased, the measured output voltage decreased. This inverse relationship aligns with the expected behavior of a differential amplifier, where the output signal is proportional to the difference between the inverting and non-inverting inputs, multiplied by the amplifier's gain.
- The differential amplifier circuit is a highly versatile op-amp configuration. By adding additional resistors in parallel with the input resistors, the circuit can be modified to either “add” or “subtract” the voltages applied to the respective inputs.

Conclusions:

The op-amp differential amplifier circuit performed as expected, amplifying the voltage difference between two input signals. The measured output voltages were in close agreement with the calculated output values, with small percentage deviations observed, which are typical due to practical limitations such as component tolerances and measurement inaccuracies. Differential amplifiers have a wide range of important applications. They are widely used in instrumentation systems to amplify small differential signals while rejecting large common-mode voltages. In audio processing circuits, they improve sound quality by minimizing noise and distortion, especially in balanced audio lines. Additionally, differential amplifiers are employed in data acquisition from strain gauge sensors, temperature sensors, and pressure sensors, enhancing the accuracy and reliability of the small output signals from these devices. Overall, the results of this experiment affirm the expected theoretical behavior of the circuit while revealing opportunities for refinement in practical applications across various domains.

EXPERIMENT 5

PART A: OPERATIONAL AMPLIFIER AS A DIFFERENTIATOR

Aim - To study the application of an op-amp as differentiator..

Apparatus – IC 741, Resistors - 150 Ω , 1.5 k Ω , Capacitors – 0.1 μ F, 0.01 μ F, Oscilloscope, Signal Generator, Regulated DC power supply.

Circuit Diagram

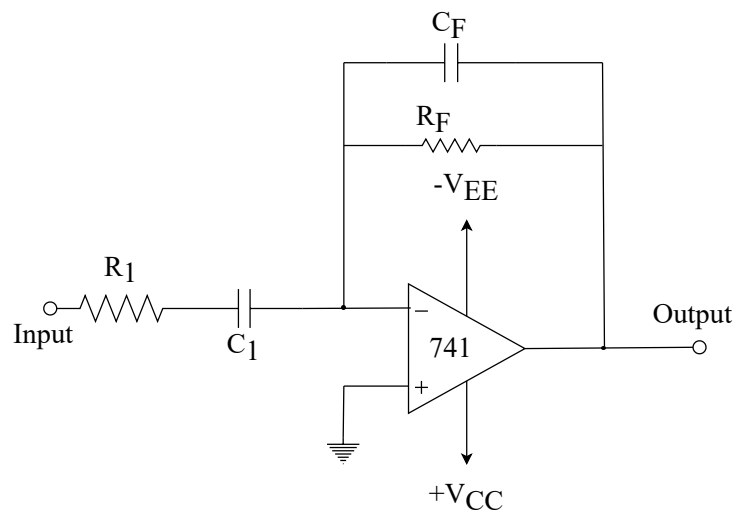


Figure 5.1: Circuit for operational amplifier as differentiator

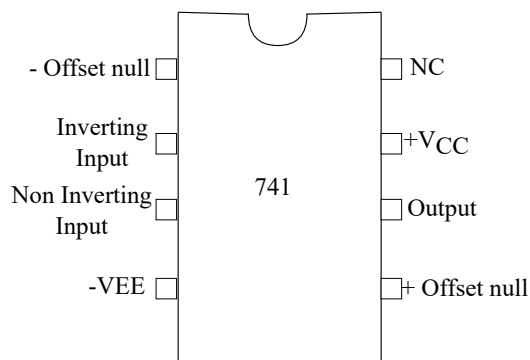


Figure 5.2: Schematic of IC 741(OP-Amp)

Design Procedure:

- Select given input frequency f_a such that,

$$f_a = \frac{1}{2\pi R_F C_1} \quad (5.38)$$

- Assume C_1 and find the value of R_F given by the above equation.

- Select $f_b = 10f_a = \frac{1}{2\pi R_1 C_1}$ and find R_1 from this equation.
- Using $R_1 C_1 = R_F C_F$, find C_F .

Experimental Procedure:

- Connect the circuit as shown in Fig. 5.1
- Apply Square/Sine wave input of desired frequency and magnitude (4 V (peak to peak) and 1 kHz).
- Observe the output voltage at pin 6 of op-amp.

Observations:

The output waveform from a differentiator is observed by applying square and sine wave signals.

$C_1 = 0.1 \mu F$

Table 5.1: Differentiator Input and Output observations

Input-Square Wave		Output-Spikes		Input-Sine Wave		Output-Cosine Wave	
Amplitude (V _{P-P})	Time period (ms)	Amplitude (V _{P-P})	Time period (ms)	Amplitude (V _{P-P})	Time period (ms)	Amplitude (V _{P-P})	Time period (ms)
0.50	1	2.4	1	2.04	1	1.40	1
1.04	1	4.4	1	2.56	1	1.76	1
1.56	1	6.4	1	3.04	1	2.08	1
2.04	1	8.8	1	3.52	1	2.48	1
2.50	1	10.8	1	4.00	1	2.80	1

Sample Calculation:

- For $T = 1$ ms, $f_a = \frac{1}{T} = 1$ kHz, where T is the time period of input signal in seconds.
- Select input capacitance $C_1 = 0.1 \mu F$. From the equation of input frequency Eq. 5.38, the feedback resistance R_F can be calculated as

$$R_F = \frac{1}{2\pi f_a C_1} = \frac{1}{2\pi \times 1000 \times 0.1 \times 10^{-6}} = 1591.55 \Omega \quad (5.39)$$

- Input capacitance $C_1 = 0.1 \mu F$. From the equation $f_b = 10f_a = \frac{1}{2\pi R_1 C_1}$, Input resistance can be calculated as

$$R_1 = \frac{1}{2\pi \times 10 f_a C_1} = \frac{1}{2\pi \times 10 \times 1000 \times 0.1 \times 10^{-6}} = 159.15 \Omega \quad (5.40)$$

- From the relation $R_1 C_1 = R_F C_F$, C_F can be calculated as

$$C_F = \frac{R_1 C_1}{R_F} = \frac{159.15 \times 0.1 \times 10^{-6}}{1591.55} = 0.01 \mu F \quad (5.41)$$

Results and discussions:

- The experiment successfully demonstrated the operation of an op-amp (IC 741) as a differentiator. For a square-wave input with a peak-to-peak amplitude of 0.5 V, the output showed spikes with an amplitude of 2.4 V and a time period of 1 ms. For a sine-wave input with a peak-to-peak amplitude of 2.04 V, the output displayed a cosine-like waveform with a peak-to-peak amplitude of 2.08 V and a time period of 1 ms. These results, obtained using the IC 741 operational amplifier, were consistent across varying input amplitudes, confirming the differentiating behavior of the circuit.

- The output characteristics demonstrated that the differentiator effectively amplifies the rate of change of the input signals, producing significant outputs in response to varying input amplitudes. The output spikes for square wave inputs reflected the abrupt changes, while the output cosine waves for sine wave inputs showcased the differentiator's ability to generate phase-shifted signals proportional to the input's derivative.
- Fig. 5.5 shows the standard output of op-amp as differentiator.

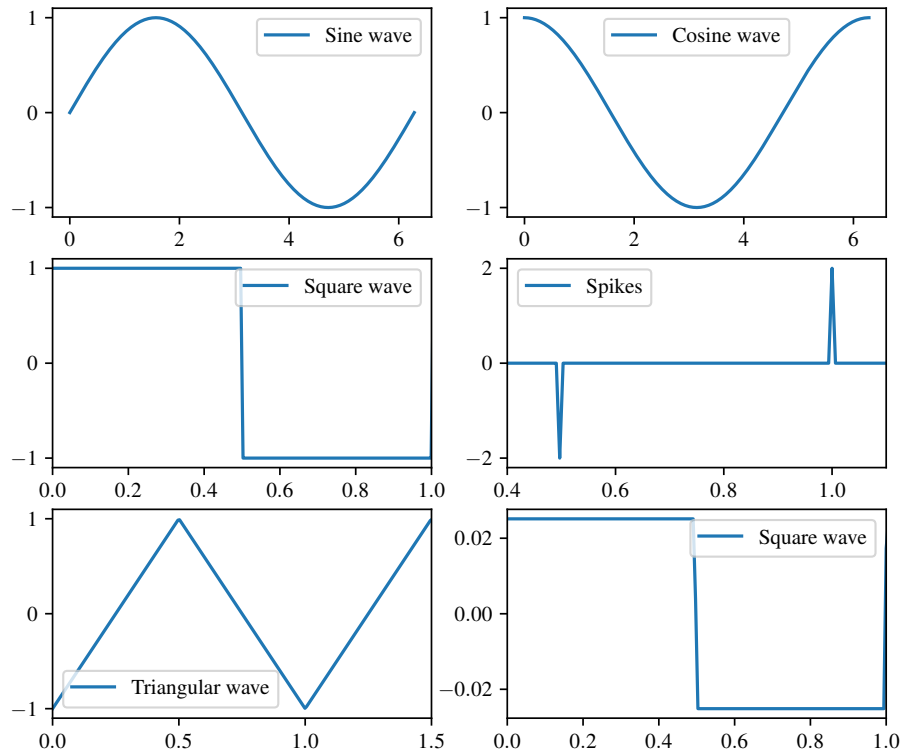


Figure 5.3: Standard output obtained from op-amp as differentiator

Conclusions:

The operation of the operational amplifier (op-amp) configured as a differentiator was thoroughly explored in this study. The output signal generated by the differentiator is directly proportional to the rate of change of the input signal, effectively representing the time derivative of the input waveform. This allows the circuit to perform the mathematical operation of differentiation. The use of an IC 741 operational amplifier, along with appropriate component values, allowed for accurate differentiation, with the output waveforms demonstrating the expected behavior in both amplitude and time period. The differentiator's ability to respond to rapid changes in the input signal is particularly significant, as it provides an amplified output that accurately reflects these changes. For instance, sharp increases or decreases in the input result in correspondingly large outputs, which is crucial for applications requiring precise measurements of changes over time, such as in signal processing and control systems. Moreover, this configuration can achieve a constant output for steadily varying input signals. When the input varies at a consistent rate, like in a linear ramp signal, the differentiator maintains a stable output, showcasing its utility in systems that require stability and reliability even with dynamic input conditions. In summary, the op-amp differentiator serves as a powerful tool for implementing mathematical differentiation while offering practical advantages in maintaining constant outputs for steadily varying signals (smooth transition over time). These capabilities make it a valuable component in various applications, including automatic control systems, analog computing, and dynamic signal processing.

EXPERIMENT 5

PART B: OPERATIONAL AMPLIFIER AS AN INTEGRATOR

Aim - To study the application of an op-amp as an integrator.

Apparatus – IC 741, Resistors - 150 Ω , 1.5 k Ω , Capacitors – 0.1 μ F, Oscilloscope, Signal Generator, Regulated DC power supply 0-30 V.

Circuit Diagram

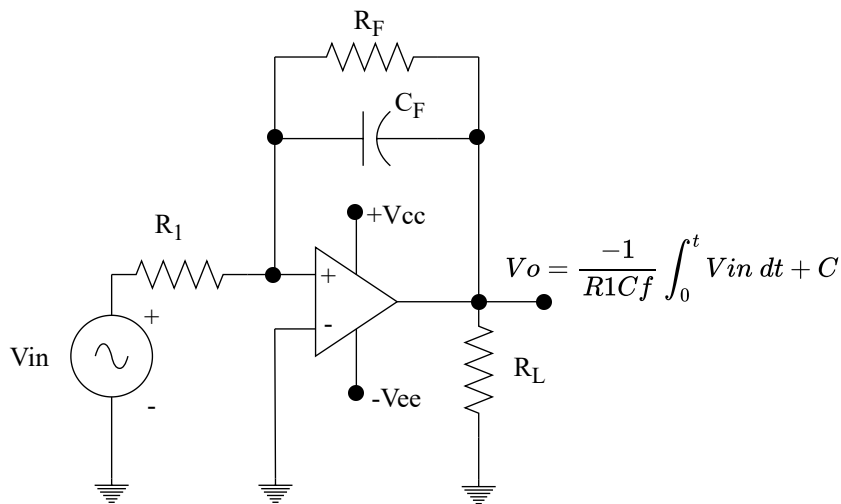


Figure 5.4: Circuit for operational amplifier as integrator

Design Procedure:

- Select given input frequency f_a such that

$$f_a = \frac{1}{2\pi R_F C_F} \quad (5.42)$$

- Assume C_F and find R_F using above relation
- Select $R_F = 10R_1$, find R_1 from this equation.
- The peak to peak output voltage is given by

$$V_o = -\frac{1}{R_1 C_f} \int_0^t V_{in} dt + C \quad (5.43)$$

Experimental Procedure:

- Connect the circuit as shown in the circuit diagram shown in Fig. 5.4.
- Apply Square/Sine wave input of desired frequency and magnitude 4V (peak to peak and 1 kHz).
- Observe the output voltage at pin 6 of the op-amp.

Table 5.2: Integrator Input and Output observations

Input-Square Wave		Output-Triangular		Input-Sine Wave		Output-Cosine Wave	
Amplitude (V _{P-P})	Time period (ms)	Amplitude (V _{P-P})	Time period (ms)	Amplitude (V _{P-P})	Time period (ms)	Amplitude (V _{P-P})	Time period (ms)
0.50	0.1	3.80	0.1	0.792	0.1	4.56	0.1
1.00	0.1	6.64	0.1	1.170	0.1	6.84	0.1
1.50	0.1	10.8	0.1	1.600	0.1	9.00	0.1
2.00	0.1	13.8	0.1	2.020	0.1	11.4	0.1
2.50	0.1	18.9	0.1	2.380	0.1	13.8	0.1

Sample calculations:

- For $T = 0.1$ msec, $f_a = \frac{1}{T} = 10$ kHz.
- Assuming $C_F = 0.01\mu\text{F}$, the feedback resistance R_F can be calculated by

$$R_F = \frac{1}{2\pi f_a C_F} = \frac{1}{2\pi \times 10000 \times 0.01 \times 10^{-6}} = 1591.5 \Omega \quad (5.44)$$

- From the equation $R_F = 10R_1$, R_1 can be calculated as $1591.5/10 = 159.15\Omega$

Results and discussions:

- The experiment successfully demonstrated the operation of an operational amplifier (IC 741) configured as an integrator. The input square-wave signal with varying peak-to-peak amplitudes (0.5 V to 2.5 V) produced triangular output waveforms, as expected from an integrator circuit. Similarly, a sine-wave input signal generated a cosine-wave output as shown in Fig. 5.5, which is consistent with the theoretical behavior of an integrator. For both input signals, the time period of the output waveforms remained consistent at approximately 0.10 ms, while the peak-to-peak amplitudes of the output waveforms increased with increasing input amplitudes. This was observed for both square-wave and sine-wave input.
- The results align with the theoretical behavior of an op-amp integrator. An integrator circuit produces an output voltage that is directly proportional to the integral of the input signal. This means that the op-amp accumulates the input signal over time, leading to a gradually changing output. For square-wave inputs, the integrator output was a triangular waveform, which is expected since the integral of a square wave is a ramp signal with sharp transitions between positive and negative slopes. Similarly, for sine-wave inputs, the output waveform was a cosine wave, reflecting the mathematical relationship between sine and cosine functions during integration.
- The output peak-to-peak voltage increased as the input amplitude increased, confirming that the output voltage accumulates the input signal's value over time. This behavior is characteristic of an integrator, which integrates the input voltage with respect to time, leading to a ramp or varying output voltage. The time period of the output remained constant across different input signal amplitudes, as expected for this configuration.

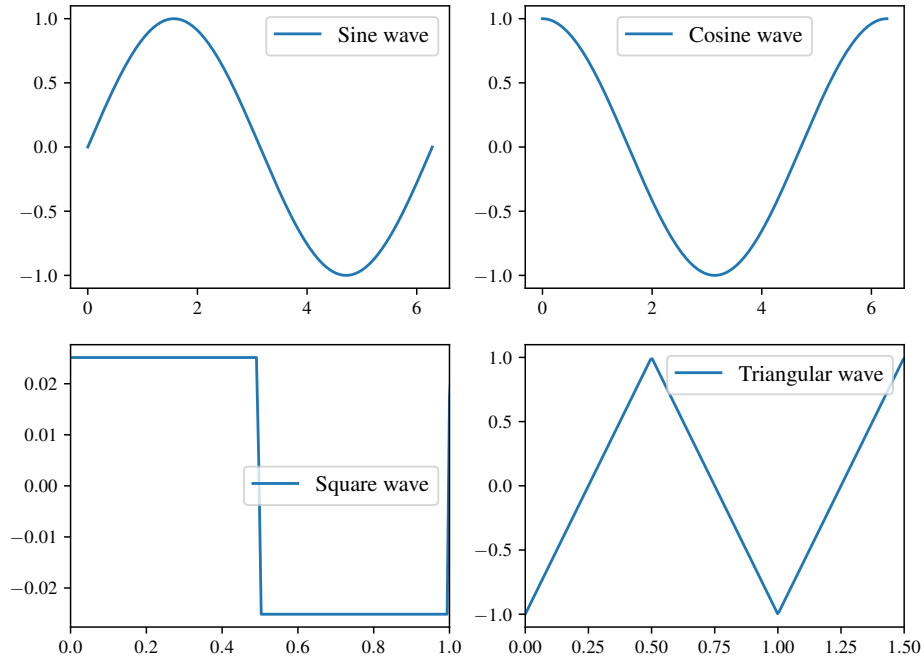


Figure 5.5: Standard output obtained from op-amp as differentiator

Conclusions:

The operation of the op-amp integrator was thoroughly explored in this experiment, confirming that the output signal is directly proportional to the negative time integral of the input signal. This means the output represents the area under the input waveform, effectively simulating the mathematical operation of integration. The output waveforms for square-wave and sine-wave inputs followed the expected patterns, with triangular and cosine waves, respectively. The output peak-to-peak amplitude increased in proportion to the input amplitude, confirming the integrator's function of accumulating the input signal's values over time. This experiment highlights the importance of the op-amp integrator in generating steadily varying signals, such as ramp signals, from constant inputs. This ability to produce smooth transitions in output is useful in a variety of applications, including analog signal processing, control systems, and automation, where dynamic control voltages are required. The op-amp integrator thus serves as a vital component in numerous systems where integration is a key operation, offering a practical solution for generating ramp signals, accumulating data, and processing time-dependent information.